

University of Wollongong

CSCI322

Assignment 2 (Due 12/11/2026)

Task 1: DNS Deployment and Configuration (4 marks)

Create the DNS zone file for the domain techcollege.edu.au and its reverse zone. The domain uses the network 172.16.20.0/24.

- Set up the DNS server on ns1.techcollege.edu.au.
- Use SOA serial numbers in the format yyymmddss.
- Add one MX record with priority 20 pointing to mail.techcollege.edu.au.
- Create suitable A and PTR records for:
 - app1.techcollege.edu.au (172.16.20.10)
 - app2.techcollege.edu.au (172.16.20.11)
 - lab1.techcollege.edu.au (172.16.20.30)
 - lab2.techcollege.edu.au (172.16.20.31)
- Add CNAME records so that www points to app2, dns points to ns1, and mail points to app1.

Task 2: Enterprise Network Design (6 marks)

A company operates a headquarters and a branch office. The Internet gateway at headquarters has IP address 172.20.1.1/24. The headquarters network connects to the Internet through RouterHQ. The branch office connects to headquarters through RouterBR. The headquarters supports up to 8,000 devices and the branch office supports up to 500 devices.

- Using the private block 10.10.0.0/16, propose suitable subnet ranges for both sites and show your calculations.
- Draw a network diagram showing all router interfaces with CIDR notation. Label interfaces as eth0 and eth1.
- Show the routing table of RouterHQ with the columns Destination, Gateway, Genmask, Flags and Iface as displayed by the Linux route command.

Submission

Submit a ZIP file named a2_yourUserID.zip containing the DNS zone files and answer.pdf with your solutions for Task 2. Implementation of these two tasks on virtual machines is not required.